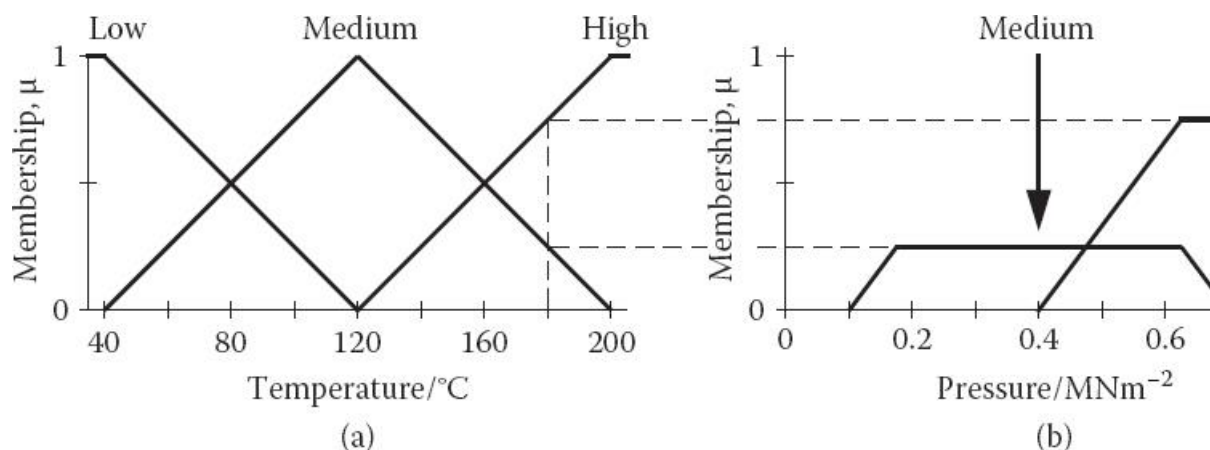


1. Prove the fuzzy DeMorgan laws:
 - i. $A \cap B = (A^c \cup B^c)^c$
 - ii. $A \cup B = (A^c \cap B^c)^c$
2. Consider a rule base that contains the following fuzzy rules:
 - if temperature is high then pressure becomes high.*
 - if temperature is medium then pressure becomes medium.*
 - if temperature is low then pressure becomes low.*

Membership functions for pressure derived from these rules for a temperature of 180°C is depicted below.



How will you proceed from here to obtain a defuzzified value as output.

3. *Premature Convergence* and *Stalled Evolution* are two of the main selection pitfalls in genetic algorithms. Briefly explain.
4. Briefly explain the basic foraging behaviour of real ants that is modelled to create the artificial ant colony systems for solving optimization problems.
5. Briefly describe the influence of *swarm size*, *number of iterations* and *acceleration coefficients* on the basic Particle Swarm Optimization (PSO) algorithm.
6. Describe the commonly used fuzzy membership functions.
7. Let a Mamdani model contains the following three rules:
 - a. *IF x is Small THEN y is Medium.*
 - b. *IF x is Medium THEN y is Large.*
 - c. *IF x is Large THEN y is Small.*

where the fuzzy sets Small, Medium, and Large are defined as:

$$\text{Small} = 1.0/1 + 0.5/2$$

$$\text{Medium} = 0.5/2 + 1.0/3 + 0.6/4 + 0.3/5$$

$$\text{Large} = 0.4/4 + 0.7/5 + 1.0/6$$

If x is X , where X is a fuzzy set (on the same universe of discourse as the above three fuzzy sets) defined as $X = 0.3/3 + 1.0/4 + 0.5/6$, is the input to the model, use the max-min compositional rule of inference to compute the output before defuzzification.

8. Initial population of a genetic algorithm implementation for solving a problem is shown in the table below. The outcome of applying the selection operator on the initial population is also shown.

String	Fitness	$p(s_i)$ probability of selection of string = $f_i/\sum f$	Expected count = $f_i/\bar{f}$	Actual count (via roulette wheel selection)
S_1 : 1010	3	3/9	$3/(9/4) = 4/3$	1
S_2 : 0000	1	1/9	$1/(9/4) = 4/9$	0
S_3 : 1101	2	2/9	$2/(9/4) = 8/9$	1
S_4 : 0110	3	3/9	$3/(9/4) = 4/3$	2

- i. Give an explanation of the entries shown in the above table.
 - ii. If one point crossover method is used, derive two possible populations for the next generation. Assume that the crossover and mutation probabilities are 0.9 and 0.1, respectively.
9. Illustrate how the ACO metaheuristic operates by describing how it has been applied to the Travelling salesman problem or any other combinatorial optimization problem.
10. Illustrate the first two rounds of using the PSO algorithm to find the minimum of the function $f(x,y) = (x + 2y - 7)^2 + (2x + y - 5)^2$. Use 4 particles with initial positions (7,1), (-1,9), (-10,1) and (-2,-5). The inertia, cognitive and social constants may be set to 1.
11. How does each of the following three categories of algorithms attempt to balance between exploitation and exploration of the solution space:
- i. Genetic Algorithms
 - ii. Ant Colony Optimization Algorithms
 - iii. Particle Swarm Optimization Algorithms
12. Consider the following hyperparameter settings for a Genetic Algorithm implementation for solving the Travelling Salesman Problem:

NUMCITIES	= 29
EliteNumber	= 2
ToursPerGeneration	= 100
NumberGenerations	= 50000
TournamentNumber	= 4
ProbCrossover	= 0.95
ProbMutation	= 0.25

State the implications of each of these parameters on the search process for finding an acceptable solution.

13. Describe how fuzzy sets could be applied to deal with the vagueness in the meaning of the following phrases:

- Water level is low.
- Temperature is high.
- Pressure is high.

14. What is meant by the term defuzzification? Briefly describe the most common method of defuzzification?

15. What is the significance of mutation in a genetic algorithm? What could happen if the mutation probability is set to zero?

16. If we were to rate the most intelligent creatures on this planet, it is doubtful that ants would be on anyone's "Top Ten List." And yet, ant colonies exhibit remarkable intelligence. Provide a brief explanation justifying this observation regarding the collective intelligence exhibited by ant colonies.

17. Briefly describe the basic Particle Swarm Optimization (PSO) algorithm.

18. Explain the commonly used fuzzy membership functions.

19. Illustrate with an example the functioning of a *fuzzy inference system*.

20. Describe the life cycle of a genetic algorithm.

21. Imagine that we are visiting a carnival that has many attractions to experience. Each attraction is located in a different area, with varying distances between attractions. The problem to be solved is finding an optimal path between all the attractions. Show how an ant colony optimization algorithm can be configured to solve this Problem.

22. Illustrate with an example how the PSO algorithm is used to find an approximate solution to an optimization problem.

23. Explain the pros and cons of the following nature-inspired algorithms.

- iv. Genetic Algorithms
- v. Ant Colony Optimization Algorithms
- vi. Particle Swarm Optimization Algorithms

24. Describe the important hyper parameters of the genetic, ACO and PSO algorithms and the impact of each of them on the performance and effectiveness of the respective algorithms.

25. Develop a membership function for "Tall" and derive from it the membership function for "Very Tall". Explain the mechanism used.

26. *Premature Convergence* and *Stalled Evolution* are two of the main selection pitfalls in genetic algorithms. Briefly explain

27. Distinguish between *Strong* and *Weak* Selection methods in genetic algorithms.

28. Briefly explain why fitness-proportionate selection is prone to both premature convergence and stalled evolution.
29. Write down the pseudocode for implementing *roulette wheel selection with replacement* (fitness-proportional selection).
30. Write down the pseudocode for implementing *stochastic universal selection*.
31. Both *fitness-proportional selection* and its variant *stochastic universal selection* suffer from the risk of premature convergence and, later on, stalled evolution. How do fitness-scaling techniques counter both problems?
32. *Elitism can be thought of as the allowance of cloning alongside reproduction*. Justify.
33. Briefly discuss the features of swarm intelligence. (information gathering, sharing, evaluation, consensus building, execution or implementation)
34. A hackathon is an example of swarm methodology. Is it true? Justify your answer.
35. What do you mean by elitism in Genetic Algorithms? What is its objective?
36. How does Fitness scaling counters premature convergence and stalled evolution in Genetic Algorithms?
37. Briefly describe *multiobjective optimization* and *pareto optimal* set.
38. "Fuzzy logic, sometimes called possibility theory, addresses a different source of uncertainty, namely, vagueness in the use of language." Briefly explain the meaning of this statement.
39. Briefly explain the so-called *accelerated particle swarm optimization* (APSO) algorithm.
40. The intelligence exhibited by ant colonies is an example of **emergent behaviour**, unforeseen behaviour at one level that arises from a lower level. Provide an explanation of this statement.